

Amy Deng

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Fullstack Developer | Sydney

SUMMARY

Backend-focused Full-Stack Engineer specializing in building scalable web applications with Java (Spring Boot) and modern frontend frameworks (Angular, React, Vue). Proven ability in API design, asynchronous processing, and delivering end-to-end solutions from backend services to responsive user interfaces.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, GraphQL
Frameworks: Spring Boot, Spring Security, Angular, React, Vue
Messaging & Streaming: Kafka, Redis Streams, MQ
Databases: MySQL, PostgreSQL
Tools: Docker, Git, JIRA, Jenkins, Elastic Search
Concepts: Microservices, Event-driven Architecture, Outbox Pattern, Idempotency, High Concurrency

WORK EXPERIENCE

ANP Technology Pty Ltd

Mar 2024 - Present

Java Engineer

Sydney

Payment & Risk Control Platform (Workflow + Limit Management)

Backend-focused Full-Stack Engineer building a multi-tenant payment and risk-control platform, combining workflow orchestration with configurable limit management to support automated decisioning and secure transaction processing.

- Designed and developed core backend modules (paymentService, paymentAdmin, shared services) using Java (17–25, Spring Boot), following DDD and Hexagonal Architecture to improve modularity and scalability
- Built a configurable limit management system supporting multi-dimensional rules (amount thresholds, operating hours, channel constraints) for tenant-level (house) control
- Implemented RBAC with tenant-aware data isolation, enforcing both functional and data-level access control across services
- Developed event-driven components using the Outbox pattern and Redis Streams, improving reliability and enabling asynchronous task processing
- Designed lease-based scheduling for distributed workflow execution, reducing duplicate processing and improving coordination across services
- Optimized MySQL and Redis interactions, improving system performance and stability under concurrent workloads
- Maintained database schemas, API documentation, and test coverage (H2-based integration tests), ensuring consistency and faster team onboarding

Independent Development

Aug 2023 - Mar 2024

Fullstack developer

Sydney

Event-Driven Payment Platform

Tech: Java, Spring Boot, Kafka, Redis, MySQL, Docker

- Designed and implemented an event-driven microservices architecture for order and payment processing
- Applied Outbox Pattern to ensure consistency between database transactions and message publishing
- Built idempotent consumers to handle duplicate message delivery
- Implemented retry mechanisms and Dead Letter Queue (DLQ) for failure handling
- Integrated OAuth2 authentication for secure API access
- Containerized services using Docker Compose

Flash Sale System (High-Concurrency System)

Tech: Spring Boot, Redis, Kafka, MySQL

- Designed a high-concurrency system to handle traffic spikes during flash sales
- Prevented overselling using Redis and Lua scripts for atomic inventory operations
- Used Kafka to decouple request handling from order processing
- Implemented rate limiting using token bucket algorithm
- Improved system throughput and resilience under high load

Workflow Engine Platform

Tech: Spring Boot, Camunda, MySQL, Angular

- Developed a workflow platform supporting BPMN-based business processes
- Designed multi-step approval workflows such as expense approval systems
- Integrated workflow engine for process orchestration and task management
- Built RESTful APIs for workflow operations
- Implemented asynchronous jobs and timer-based events

Articulate Pty Ltd

Oct 2021 - Aug 2023

Java Engineer

Sydney, Australia

Integrated Supplier Platform for Online Gaming

Full-Stack Engineer contributing to a high-traffic platform aggregating multiple gaming suppliers and exposing unified APIs to downstream clients. Developed multi-account support, reporting features, and seamless integration across diverse suppliers.

- Designed and implemented backend services using Spring Boot and PostgreSQL, with Vue.js for frontend interfaces, ensuring scalable and maintainable code
- Built a dynamic log system that aggregates supplier requests, identifies failed requests, and automatically sends alerts to downstream clients for proactive monitoring
- Participated in DevOps tasks, including CI/CD pipeline setup, server deployment, and monitoring server metrics to ensure system reliability
- Collaborated with cross-functional teams to deliver new features, optimize performance, and resolve issues efficiently
- Actively involved in code reviews and design discussions, providing feedback to improve code quality and system architecture

Jiangxi JieChuang Internet of Things Technology CO. LTD.

Jan 2019 - Aug 2019

Java Developer

Nanchang, China

- Developed web services using Java Spring MVC
- Implemented database operations in MySQL
- Assisted in building automated testing scripts for REST APIs

EDUCATION

University of New South Wales

Sep 2019 - Aug 2021

Information technology Master Engineering faculty

Sydney, Australia

- Relevant courses: Web frontend programming, Database Systems, Machine Learning & Data Mining

East China JiaoTong University

Sep 2015 - Jun 2019

Software engineering Bachelor Software faculty

Nanchang, China

- Relevant courses: C++ programming, Java programming, JSP Web Programming